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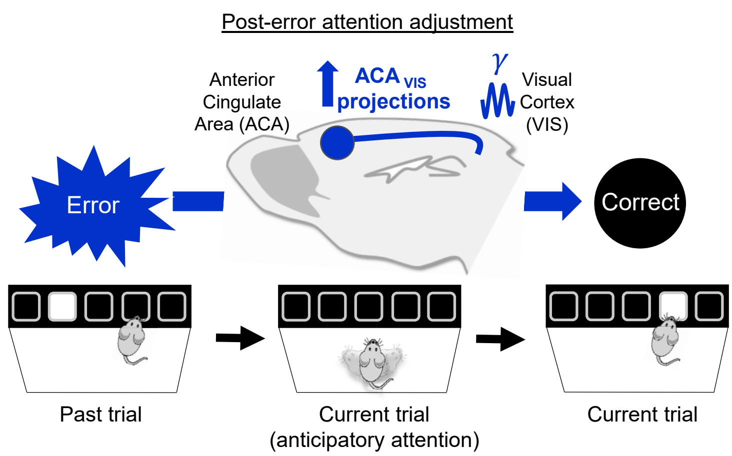
Post-error recruitment of frontal sensory cortical projections promotes attention in mice

Graphical Abstract



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In Brief

Norman et. al found that behavioral errors recruit frontal sensory projections in mice. 30-Hz optogenetic stimulation of this pathway modulates performance following errors in behaving mice and recapitulates neurophysiological hallmarks of attention in anesthetized mice. Frontal sensory projections therefore link error monitoring with attention adjustments for behavioral adaptation.

Highlights

- Top-down frontal sensory projections are selectively recruited after error trials
- 30-Hz optogenetic stimulation of top-down neurons promotes post-error performance
- Post-error performance adjustment requires anticipatory top-down activity
- 30-Hz optogenetic top-down stimulation promotes a hallmark feature of attention

Norman et al., 2021, Neuron *109*, 1202–1213
April 7, 2021
https://doi.org/10.1016/j.neuron.2021.02.001

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Article

Post-error recruitment of frontal sensory cortical projections promotes attention in mice

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https://doi.org/10.1016/j.neuron.2021.02.001

SUMMARY

The frontal cortex, especially the anterior cingulate cortex area (ACA), is essential for exerting cognitive control after errors, but the mechanisms that enable modulation of attention to improve performance after errors are poorly understood. Here we demonstrate that during a mouse visual attention task, ACA neurons projecting to the visual cortex (VIS; ACA_{VIS} neurons) are recruited selectively by recent errors. Optogenetic manipulations of this pathway collectively support the model that rhythmic modulation of ACA_{VIS} neurons in anticipation of visual stimuli is crucial for adjusting performance following errors. 30-Hz optogenetic stimulation of ACA_{VIS} neurons in anesthetized mice recapitulates the increased gamma and reduced theta VIS oscillatory changes that are associated with endogenous post-error performance during behavior and subsequently increased visually evoked spiking, a hallmark feature of visual attention. This frontal sensory neural circuit links error monitoring with implementing adjustments of attention to guide behavioral adaptation, pointing to a circuit-based mechanism for promoting cognitive control.

INTRODUCTION

Adaptive goal-directed behavior requires cognitive control—systems that monitor contextually relevant internal states and external events and implement strategic adjustments in information processing, learning, and behavior after errors are made (Botvinick et al., 2004; Ullsperger et al., 2014). In people, non-human primates, and rodents, the medial frontal cortex is essential for cognitive control (Ridderinkhof et al., 2004; Shenhav et al., 2016). Notably, distinct populations of neurons in the medial frontal cortex monitor specific internal and external conditions ranging from the history of positive (e.g., rewards) and negative (e.g., error or failure) outcomes, conflicts, and uncertainty (Matsumoto et al., 2007; Ridderinkhof et al., 2004; Rushworth and Behrens, 2008). Although the neural mechanisms of performance monitoring are well characterized, the circuit mechanisms that implement subsequent cognitive control are unclear. Cognitive control is aided by attentional processes that enhance our ability to detect relevant sensory stimuli. These highly dynamic processes must be deployed efficiently and flexibly to

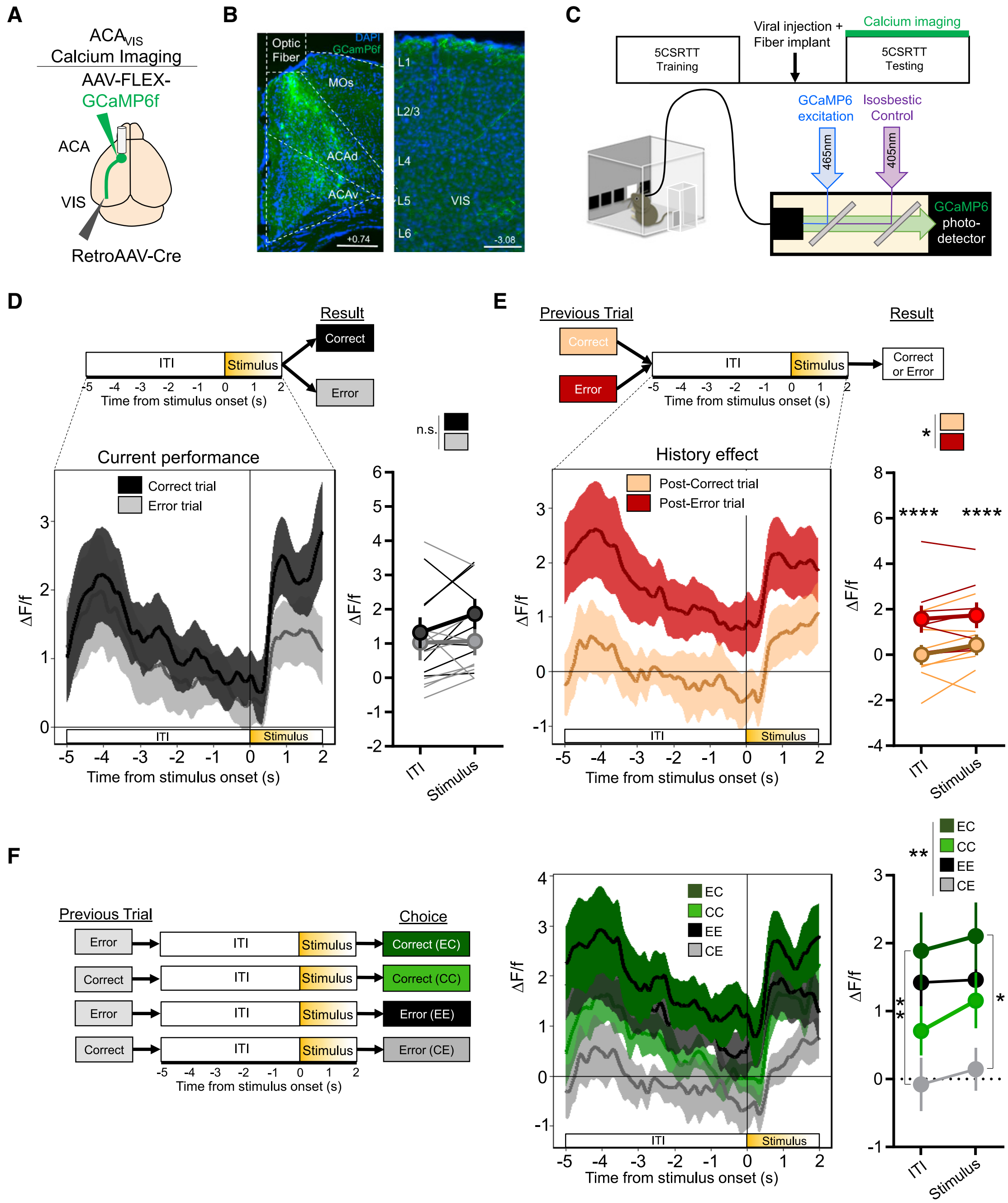

Figure 1. ACA$_{VIS}$ neurons are selectively recruited following a recent history of error trials

(A) Intersectional viral strategy. To selectively express the GCaMP6f calcium indicator in ACA$_{VIS}$ neurons, Cre-dependent GCaMP6f and retrograde Cre-encoding AAVs were injected bilaterally into the ACA and visual cortex (VIS), respectively. An optic fiber was implanted at the ACA for calcium-dependent recordings of ACA$_{VIS}$ neurons using fiber photometry.

(B) Representative images of GCaMP6f (green)-expressing ACA$_{VIS}$ neurons, DAPI (blue), and the optic fiber in the frontal cortex (left; scale bar, 200 μm) and axon terminals in the VIS (right; scale bar, 100 μm).

(C) Mice (n = 8 mice) were trained on the 5CSRTT before virus injection. After 3 weeks to allow maximal virus expression, fiber photometry recordings of ACA$_{VIS}$ neurons were conducted while mice performed 2 days of 5CSRTT (8 mice, 1,024 total trials) with a fixed 5-s intertrial interval (ITI) and pseudorandomized stimulus duration (2.0, 1.5, 1.0, or 0.8 s).

(D) During the 5CSRTT, there was no difference in averaged ACA$_{VIS}$ neuron activity between correct and error trials (two-way repeated measures [RM] ANOVA, $F_{1,7} = 3.592$, p = 0.0999, n = 8 mice) during the ITI (−5:0 s) or during the stimulus period (0:2 s).

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improve performance (Fiebelkorn and Kastner, 2020). The circuit mechanisms that link cognitive control with attention following errors are poorly understood.

One key circuit conserved across species for brain-wide attention consists of direct top-down cortico-cortical projections from the frontal cortex to sensory cortical areas (Knudsen, 2007; Moore and Zirnsak, 2017). Frontal projections to visual areas are important for attention modulation of visual processing in primates (Moore and Zirnsak, 2017). In mice, frontal sensory projection neurons from the anterior cingulate area (ACA) to the visual cortex (VIS; ACA_{VIS} neurons) regulate the ability to detect visual features—a hallmark feature of visual attention (Zhang et al., 2014). However, recent studies report mixed contributions of frontal sensory circuits to attention (Fiebelkorn and Kastner, 2020; Krauzlis et al., 2013), suggesting that ACA_{VIS} neurons may be recruited only under particular conditions to mediate attention (Fiebelkorn and Kastner, 2020). We tested this hypothesis by selectively monitoring and manipulating neural activity in ACA_{VIS} neurons projecting to the VIS in mice performing an attention task. The 5-choice serial reaction time task (5CSRTT) (Carli et al., 1983; Figure S1A) requires mice to sustain and divide their attention across five response windows in anticipation of a random presentation of a brief stimulus at one of the five locations. This self-paced, free-moving, attention-demanding task provides a holistic evaluation of a dynamic range of factors that underlie attention in rodents (Dalley et al., 2004; Koike et al., 2016; Lustig et al., 2013; Mar et al., 2013).

RESULTS

ACA_{VIS} neurons are recruited following errors

We first characterized the time course and circumstances of ACA_{VIS} neuron recruitment during performance in the 5CSRTT. The 5CSRTT assays were conducted in automated, standardized touch-screen operant chambers that required mice to maintain divided attention among five areas. A brief visual stimulus of a solid white square was presented randomly in one of the five locations on a black flat touch screen, and a nose poke to that location was rewarded (Bari et al., 2008). The number of trials with touches to the screen that flashed compared with missed and incorrect touches was the primary operational definition of attention. Together with these primary measures, other 5CSRTT behavioral measures assessed relatively independent indices of cognition, including, but not limited to, reward collect latency and premature and preservative responses (Figure S1; Robbins, 2002). After mice completed four of five daily training sessions with 80% accuracy or higher, 20% or less omissions, and at least 50 trials with a 2-s stimulus duration (Figure S1), the activity of ACA_{VIS} neurons expressing the calcium indicator GCaMP6f was assessed using fiber photometry as mice performed the 5CSRTT (Figures 1A–1C and S2A).

We found that ACA_{VIS} neuron activity in single trials did not differ between correct choices, when attention was properly allocated, and errors (incorrect choice or omission) (Figures 1D and S2B). Rather, ACA_{VIS} neuron activity depended on the outcome of the previous trial, showing elevated activity during the intertrial interval (ITI) and stimulus presentation in trials immediately following an error compared with a correct response (Figure 1E). This result suggests that ACA_{VIS} neuron activity provides an error-monitoring signal that is engaged after an error occurs. Investigating other behavioral factors that could, in principle, recruit ACA_{VIS} neurons revealed no differences in ACA_{VIS} neuron activity between correct or incorrect trials based on response latency or reward collection latency (Figures S2C–S2E), demonstrating that ACA_{VIS} neurons were not differentially recruited by operations that vary with processing speed, motor preparation, or motivation.

To determine whether overall ACA_{VIS} activity correlated with improved performance after an error was made, we classified trials into four groups based on the combined outcome of the previous and current trials (Figure 1F). Although trial combinations differed significantly overall, activity in correct trials after an error (EC) did not differ from activity in consecutive error trials (EE) during the ITI or stimulus presentation (Figure 1F). Furthermore, ACA_{VIS} neurons responded to the visual stimulus presentation during EC, EE, and CC trials but not during CE trials (Figure S2F), suggesting that ACA_{VIS} neuron activity reflects the complex combination of response to past error and ongoing level of anticipatory attention. Therefore, increased ACA_{VIS} neuron activity upon visual stimulation presentation was not always correlated with performance during the current trial. The data show graded recruitment of ACA_{VIS} neurons during the visual attention task that was greatest directly after an error.

Rhythmic optogenetic stimulation of ACA_{VIS} neurons restores task performance after errors

Fiber photometry measures sum neural activity and cannot detect finer temporal patterns of activity, such as oscillations, that may be crucial for error correction signals from ACA_{VIS} neurons. We therefore used optogenetic stimulation methods to investigate the extent to which specific temporal patterns of activity contribute to post-error performance. Previous studies report that prefrontal cortex (PFC) oscillatory activity at low gamma frequencies (30–40 Hz) increase during the ITI of correct trials, whereas lower frequencies (5–10 Hz) accompany omissions. Moreover, optogenetic stimulation of PFC parvalbumin (PV) interneurons at these frequencies alters attention task performance in mice (Kim et al., 2016). Because PV interneurons

(E) Trials following an error showed greater ACA_{VIS} activity than trials following correct responses (two-way RM ANOVA, $F_{1,7}$ = 2.561, *p = 0.0244, Sidak multiple comparisons at the ITI and stimulus period, ****p < 0.0001, ****p < 0.0001, n = 8 mice).

(F) Left: Four possible trial types based on previous and current trial results. Right: there was a significant difference between trial types (two-way RM ANOVA, $F_{3,21}$ = 7.632, **p = 0.0026, n = 8 mice, Tukey's multiple comparisons; ITI: EC versus EE, p = 0.7282; EC versus EE, p = 0.09326; EC versus CE, **p = 0.0095; CC versus CE, p = 0.0515; EE versus CE, p = 0.0592; Stimulus: EC versus EE, p = 0.6333; EC versus CC, p = 0.2368; EC versus CE, *p = 0.0374; CC versus EE, p = 0.9626; CC versus CE, p = 0.2602; EE versus CE, p = 0.0753).

Error bars indicate mean ± SEM; n.s., non-significant. *p < 0.05, **p < 0.01, ****p < 0.0001. See also Figure S2.

connect extensively with and modulate the activity of pyramidal neurons, these findings led us to compare the effects of delivering 5-Hz and 30-Hz optical stimulation to channelrhodopsin (ChR2)-expressing ACA_{VIS} neurons in mice performing the 5CSRTT (Figures 2A–2C).

Although adult mice typically performed worse in the 5CSRTT after error trials (Figure S3), delivering 30-Hz blue light illumination to ACA_{VIS} neurons prior to stimulus presentation significantly improved performance after error trials by increasing accuracy and reducing omissions, well-established indicators of improved attention (Figures 2D, 2E, and S4A–S4E). When the previous trial was correct, however, 30-Hz blue light illumination did not affect performance (Figures 2D and 2E). The post-error behavioral improvement with optogenetic modulation was specific to 30-Hz frequency of illumination; 5-Hz stimulation had no significant effect on performance following error or correct trials (Figures 2D and 2E). Blue light illumination of the ACA at 5 Hz or 30 Hz delivered during stimulus presentation elicited no discernable effect on 5CSRTT performance (Figures 2F, 2G, and S4J–S4L), indicating that frequency-specific ACA_{VIS} neuron activity is critical for error corrections in the period before rather than during visual stimulus presentation. Notably, neither stimulation frequency altered reward collection latency, correct response latency, premature responses, or perseverative responses. Moreover, 30-Hz blue light illumination of ACA_{VIS} neurons did not affect motivation in a progressive ratio task or motor activity in an open field test (Figures S4M and S4N). The omission rate is sensitive to attention and motivation, and every operant response depends on both to some degree. If optical stimulation of ACA_{VIS} neurons altered 5CSRTT performance by affecting motivation, then it should also alter reward collection latency and the breakpoint of responses during progressive ratio testing. The results show, however, that ACA_{VIS} neuron stimulation that selectively improved accuracy and reduced omissions did not alter established measures of motivation. Rather, 30-Hz stimulation delivered after error trials increased well-established measures of attention in the 5CSRTT but not measures of motivation, motor preparation, impulsivity, or compulsivity (Figures S4F–S4I).

To gain insight into the neural mechanisms engaged by optogenetic stimulation that modulated post-error performance, we examined the differential effect of optogenetic stimulation on visual cortical neuron spiking and local field potentials (LFPs) in anesthetized mice. We analyzed *in vivo* single-unit and LFP activity in the VIS associated with 5- or 30-Hz blue light illumination of ACA_{VIS} neurons expressing ChR2 in anesthetized mice (Figure 3A). 5-Hz and 30-Hz blue light stimulation induced an immediate and robust increase in spiking activity (Figures 3B–3E). The moderate increase in evoked-spike probability during 30-Hz light stimulation was not accompanied by depolarization block or artificially high spike rates, consistent with previous studies (Bitzenhofer et al., 2017; Cardin et al., 2009). LFPs recorded from the VIS during 30-Hz blue light illumination of ACA_{VIS} neurons increased low gamma (25–50 Hz) power and reduced theta (4–8 Hz) power (Figures 3F–3H). In contrast, 5-Hz stimulation increased power across a broad frequency range (Figures 3F–3H).

To gain further insight into the physiological significance of these LFP changes recorded in anesthetized mice, we recorded electroencephalogram (EEG) signals from the VIS while mice performed the 5CSRTT. We compared the changes between EC and EE trials during the ITI, when 30-Hz light illumination improved post-error performance. We found that post-error changes of task performance were associated with increased low gamma and decreased theta LFP power (Figures 3I–3K and S5). These data suggest that optogenetic modulation of ACA_{VIS} neurons by 30-Hz light illumination may recapitulate physiological changes in VIS LFP power associated with improved post-error performance.

Optogenetic suppression of ACA_{VIS} neuron activity disrupts post-error performance

To determine when ACA_{VIS} signals to the VIS are required for error correction independent of projection to other areas through collaterals, we selectively suppressed ACA_{VIS} axon terminal activity in the VIS at key time points during the 5CSRTT (Figures 4A–4C, S6A, and S6B). ACA_{VIS} neuron projection terminals expressing the inhibitory opsin halorhodopsin eNpHR3.0 were illuminated by a light over the VIS during behavior testing. Inhibiting the ACA_{VIS} neuron projection during the 3 s immediately before stimulus presentation (−3:0 s) impaired performance, reducing accuracy and increasing omissions in trials that followed error but not correct trials (Figures 4F, 4G, and S6C–S6F). Disrupting performance required precisely timed inhibition of ACA_{VIS} neuron activity. Shifting the illumination time by 2 s earlier in the ITI (−5:-2 s) or immediately after the ITI during stimulus presentation (0:1 s) spared post-error performance (Figures 4D, 4E, 4H, and 4I). At neither time point did ACA_{VIS} neuron terminal inhibition affect performance in trials that followed correct trials (Figures 4D–4I). ACA_{VIS} neuron terminal inhibition had no effect on reward collection latency, premature responses, or perseverative responses (Figures S6G, S6I, and S6J), and it had no effect on motivation during a progressive ratio task or motor activity during open field testing (Figures S6K and S6L). Although ACA_{VIS} terminal inhibition during visual stimulus presentation reduced correct response latency (Figure S6H), it had no effect on task performance, likely reflecting a complex neural mechanism of speed-accuracy tradeoff (Heitz and Schall, 2012). Delivering light to GFP-expressing control mice induced no behavioral effects during 5CSRTT (Figure S7). ACA_{VIS} projections are causally important for improving task performance in the period just before stimulus presentation following an error.

30-Hz optogenetic stimulation of ACA_{VIS} neuron terminals promotes visual processing and post-error performance

To further elucidate how ACA_{VIS} neurons modulate VIS activity and task performance after errors, we characterized the behavioral effects of optogenetically activating ACA_{VIS} neuron terminals by illuminating the VIS with 30-Hz blue light during 5CSRTT performance in behaving mice. We assessed the neural correlates of optical stimulation by recording activity in the VIS of anesthetized mice. Here we used Chronos, a ChR2 variant that produces reliable spiking when terminals are stimulated at high frequencies (Klapoetke et al., 2014). ACA_{VIS} neuron projection terminals virally expressing Chronos were illuminated with blue light at 5 Hz or 30 Hz over the VIS during the last 3 s of the ITI

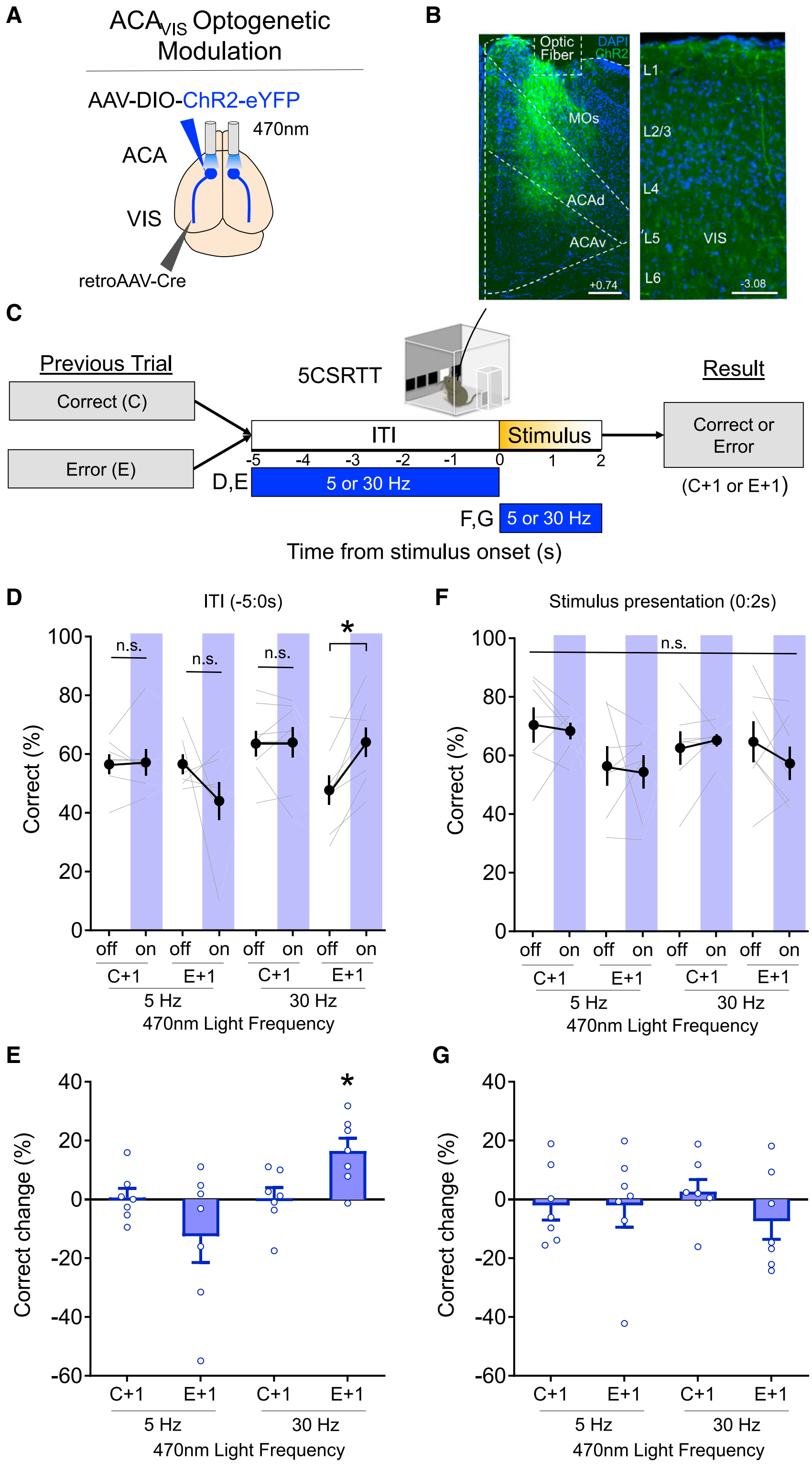

Figure 2. Optogenetic modulation of ACA$_{VIS}$ neurons selectively improves post-error performance

(A) Intersectional viral strategy. To selectively express light-gated sodium channel ChR2 in ACA$_{VIS}$ neurons, Cre-dependent ChR2 and retrograde Cre-encoding AAVs were injected bilaterally into the ACA and VIS, respectively. An optic fiber was implanted at the ACA to optically modulate the activity of ACA$_{VIS}$ neurons with temporal precision during the 5CSRTT.

(B) Representative images of ChR2 (green)-expressing ACA$_{VIS}$ neurons and DAPI (blue) in the frontal cortex (left; scale bar, 200 μm) and axon terminals in the VIS (right; scale bar, 100 μm).

(C) Experimental timeline. Mice were trained on the 5CSRTT before virus injection. After 3 weeks to allow maximal virus expression, mice performed 5CSRTT. During the 5CSRTT with a fixed 5-s ITI and pseudorandomized stimulus duration (2.0 or 1.0 s), 470-nm LED was stimulated at 5 or 30 Hz during the 5-s ITI or period of stimulus presentation of 50% of trials (7 mice, 3,337 total trials). One timing and frequency was tested per day. The effect of optogenetic stimulation on behavioral performance was compared following correct trials (C+1) and following error trials (E+1) at each tested frequency.

(D) 30-Hz blue light illumination during the ITI significantly increased correct trials (percent) after an error but not after a correct response. However, 5-Hz stimulation during the ITI had no effect on correct trials (percent) after an error or after a correct response (general linear mixed model, effect of treatment [light on versus off]: $p = 0.6467$, effect of frequency [5 versus 30 Hz]: $p = 0.0016$, planned Tukey post hoc tests: 5 Hz:C+1 light off versus on: $p = 1.000$, E+1 light off versus on: $p = 0.1963$, 30 Hz: C+1 light off versus on: $p = 1.000$, E+1 light off versus on: $^*p = 0.01410$). There was a difference in the baseline performance as mice performed worse (see also Figure S3) following errors compared with following correct trials; 30 Hz C+1 light off versus E+1 light off: $p = 0.0012$.

(E) Change in correct trials (percent light on – light off) during ITI stimulation was significantly different from zero only in the 30 Hz:E+1 group (Wilcoxon signed-rank test; 5 Hz:C+1: $p = 1.000$, 5 Hz:E+1: $p = 0.3750$, 30 Hz:C+1: $p = 0.8125$, 30 Hz:E+1: $^*p = 0.0313$).

(F) Stimulation during the period of stimulus presentation at 30 Hz or 5 Hz had no effect on correct trials (percent) after an error or after a correct response (general linear mixed model, effect of treatment [light on versus off]: $p = 0.5546$, effect of frequency [5 versus 30 Hz]: $p = 0.1779$).

(G) Change in correct trials (percent light on – light off) during the stimulus period was not different from zero in any group (Wilcoxon signed-rank test; 5 Hz:C+1: $p = 0.8125$, 5 Hz:E+1: $p = 0.8125$, 30 Hz:C+1: $p = 0.3750$, 30 Hz:E+1: $p = 0.2969$).

Error bars indicate mean ± SEM. $^*p < 0.05$. See also Figures S3 and S4.

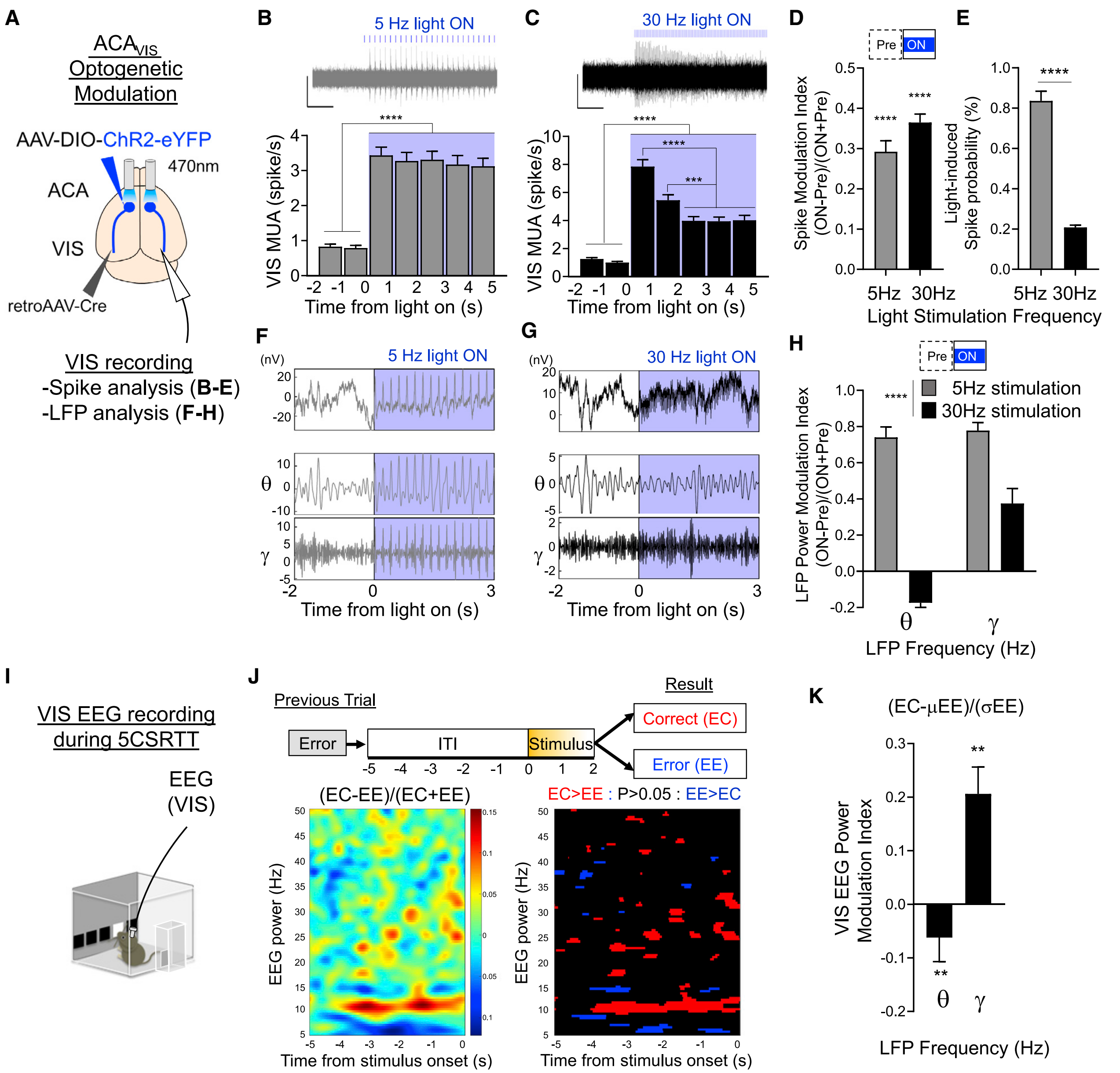

Figure 3. 30-Hz optogenetic stimulation of ACA$_{VIS}$ neurons in anesthetized mice recapitulates neural correlates of post-error performance during behavior

(A–H) VIS spike and LFP analysis of optogenetic stimulation of ACA$_{VIS}$ neurons in anesthetized mice

(A) To selectively express ChR2 in ACA$_{VIS}$ neurons, Cre-dependent ChR2 and retrograde-Cre encoding AAVs were injected bilaterally into the ACA and VIS, respectively. An optic fiber was placed at the ACA to illuminate blue light at 30 or 5 Hz, and a 16-channel silicon probe was placed in the VIS to record neuronal spiking activity (B–E) and LFP (F–H) in anesthetized mice.

(B and C) Representative traces (top) of multi-unit spike activity (MUA) in the VIS and averaged firing rate of VIS MUA before and during (B) 5Hz (n = 272 units from 3 mice) or (C) 30-Hz (n = 368 units from 3 mice) blue light stimulation (bottom). Scales: x, 1 s; y, 100 μV. MUAs are defined with a 4SD cutoff. One-way RM ANOVA: ****$p < 0.0001$, ***$p = 0.0002$.

(D) MUA during blue light illumination quantified with spike modulation index, calculated as (ON firing rate [0:5 s] − pre-firing rate [−2:0 s]) / (ON firing rate [5 s] + pre-firing rate [2 s]). Wilcoxon signed-rank test, ****$p < 0.0001$, n = 272 units (5 Hz) and 368 units (30 Hz) from 3 mice.

(E) Probability of blue light pulse-induced MUA in the VIS. n = 272 units (5 Hz) and 368 units (30 Hz) from 3 mice (unpaired t test, $t_{638} = 14.50$, ****$p < 0.0001$).

(F–H) VIS LFP analysis during optogenetic stimulation of ACA$_{VIS}$ neurons by 5-Hz and 30-Hz blue light illumination.

(F and G) Representative LFP waveforms recorded in the VIS (top) and filtered in 4- to 8-Hz (theta) and 25- to 50-Hz (gamma) bands (bottom) before and during 5-Hz (F) or 30-Hz (G) blue light stimulation.

(H) Quantification of LFP power modulation during blue light illumination by modulation index, calculated as ([ON LFP power (0:3 s) − pre-LFP power (−2:0 s)]) / ([ON LFP power (0:3 s) + pre LFP power (−2:0 s)]). Two-way RM ANOVA $F_{1,34} = 55.59$, ****$p < 0.0001$, n = 15 (5 Hz) and 21 (30 Hz) recording sites from 3 mice.

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(Figures 5A, 5B, and S8A), when ACA_{VIS} neuron activity is necessary for post-error performance (Figure 4). Stimulating ACA_{VIS} neuron terminals with 30-Hz but not 5-Hz blue light improved post-error performance (Figures 5C, 5D, and S8B–S8E) but had no effect on reward collection latency, correct response latency, premature responses, or perseverative responses. The selective effects on 5CSRTT performance are consistent with enhanced attention rather than altered motivation, motor preparation, impulsivity, or compulsivity (Figures S8F–S8I). The behavioral results were similar to the effects of stimulating ACA_{VIS} neuron "soma" in the ACA using ChR2; Chronos stimulation with 30-Hz but not 5-Hz blue light improved post-error performance, verifying that ACA_{VIS} neuron projections play a key role in post-error task performance.

Finally, in anesthetized mice, we used the same optogenetic stimulation parameters that improved behavioral performance to investigate their neurophysiological effects. An increased visually evoked spike firing rate is a key neural correlate of attention (Noudoost et al., 2010). We therefore examined the extent to which activating ACA_{VIS} projections differentially affected spiking activity evoked by visual stimuli presented after termination of blue light illumination. We confirmed that, in anesthetized mice (Figure 5E), VIS spiking activity and characteristic LFP power changes accompanied 5-Hz and 30-Hz blue light illumination of ACA_{VIS} neuron terminals expressing Chronos (Figure S9) that were similar to the effects of ChR2-based ACA_{VIS} neuron soma stimulation (Figure 3). We then analyzed the firing rate of VIS single units in response to a brief visual stimulus presentation (a white square on a black background) immediately following termination of blue light delivery in anesthetized mice. We found that visually evoked spiking in the VIS was higher after 30-Hz than after 5-Hz stimulation (Figures 5E–5G). Collectively, these findings suggest that 30-Hz optogenetic stimulation of ACA_{VIS} projections in anesthetized mice promotes subsequent VIS excitability in response to visual stimulation, a physiological effect that could contribute to improved attentional processing after errors in the 5CSRTT.

DISCUSSION

Here we showed that errors are a key internal condition for recruiting top-down signals from the frontal cortex to the VIS that support performance in an attention-guided task. Task errors trigger many behavioral adaptations, including enhanced sensory tuning to task-relevant information, that improve future outcomes (Shen et al., 2015; Totah et al., 2009; Ullsperger et al., 2014). The circuit mechanisms that link error monitoring with attention have been less clear. Our findings in mice dovetail with previous human imaging studies showing post-error activation of the frontal cortex and task-relevant visual areas (Danielmeier et al., 2011; King et al., 2010). Beyond similar activation patterns, however, our study demonstrates that direct and specific frontal sensory projections are key pathways for linking past errors to cognitive control mechanisms that promote attention. Although other frontal circuits (e.g., medial PFC PV interneurons), have been shown to modulate attention, these mechanisms are not modulated by outcome history (Kim et al., 2016). In contrast, ACA_{VIS} neuron projections provide a unique mechanism by which past behavioral errors engage frontal sensory cortical projections that support task performance. The selective engagement of these mechanisms by errors is important. Optogenetic ACA_{VIS} neuron modulation alters task performance only after error trials, not after correct trials, highlighting the functional specificity of the ACA_{VIS} neuron pathway in cognitive control-related mechanisms of attention. The selective recruitment of these top-down projections may be especially important for overcoming non-specific autonomic physiological consequences that otherwise impair performance after errors (Wessel, 2018). The ACA_{VIS} neuron pathway is not engaged after correct trials, when action correction is unnecessary and cognitive control of visual attention is unimportant for task performance.

Our optogenetic studies suggest that specific temporal patterns of activity in the ACA_{VIS} neuron pathway are critical components of post-error attention mechanisms. Thus, 30-Hz optogenetic stimulation of ACA_{VIS} neurons improved attention after errors but 5-Hz stimulation did not. *In vivo* recordings in the VIS of anesthetized mice further showed that 30-Hz optogenetic ACA_{VIS} stimulation increased the power of gamma frequency LFP oscillations in the VIS and reduced theta power. In contrast, 5-Hz optical stimulation induced a non-selective increase in LFP power across wide-ranging frequencies. Although these recordings were performed in anesthetized mice, the oscillation patterns are consistent with physiological neural correlates of improved behavioral performance and support a model that temporally coordinated activity is crucial for the brain mechanisms of post-error improvement of attention. Following errors, EEGs recorded during the anticipatory period showed higher gamma power in the VIS just before an animal made a correct choice compared with an error. Our findings from anesthetized mice that ACA_{VIS} stimulation modulated LFPs in the VIS corroborate previous studies showing that LFPs in the medial PFC (mPFC) correlate with attention performance, although prior studies did not report history-dependent effects. As mice performed a 3-choice SRTT, 30- to 40-Hz gamma power increased in the mPFC during the last 2 s of the ITI of correct trials, whereas 5- to 10-Hz power increased in omission trials (Kim et al., 2016). Moreover, 30-Hz optogenetic stimulation of

(I–K) EEG recording from the VIS during the 5CSRTT.

(I) Mice (n = 6) were trained on the 5CSRTT before being implanted with a brain surface skull screw for EEG recording of the VIS.

(J) Left: spectrogram showing the EEG power modulation index, calculated as (EC-EE)/(EC+EE) (355 EC trials, 464 EE trials). Right: t tests were calculated per time-frequency bin, and we compared power between EC and EE trials. Right: statistically significant modulation is shown as blue (EC < EE), black (non-significant), and red (EC > EE).

(K) Quantification of EEG power modulation associated with post-error performance, calculated as $(EC - \mu EE)/(\sigma EE)$ (for each EC trial, the mean power for all EE trials was subtracted and then divided by the standard deviation of all EE trials) at theta (4–8 Hz) and gamma (25–50 Hz) frequency. Wilcoxon signed-rank test, **p = 0.003 (5 Hz), **p = 0.001 (30 Hz), 355 EC trials and 464 EE trials from 6 mice. Error bars indicate mean ± SEM.

*p < 0.05, **p < 0.01, ****p < 0.0001. See also Figure S5.

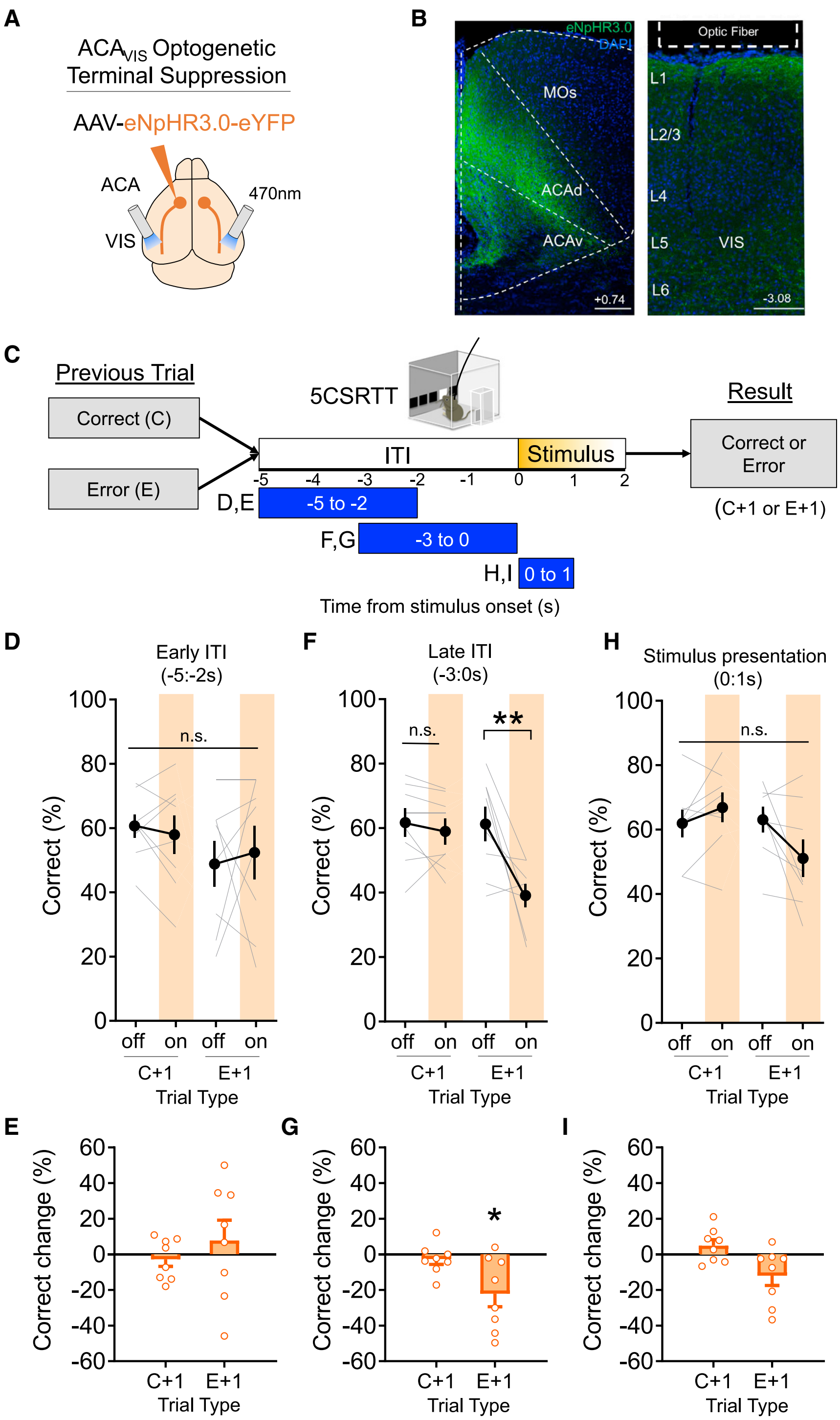

Figure 4. Optogenetic suppression of ACA$_{VIS}$ projection terminals in the VIS disrupts post-error performance

(A–I) Optogenetic suppression of ACA$_{VIS}$ neuron projection terminals during the 5CSRTT.

(A) Viral strategy. To express the light-gated chloride channel eNpHR3.0 in excitatory ACA neurons, eNpHR3.0 encoding AAV under the CaMK2 promoter was injected into the ACA. An optic fiber was implanted at the VIS to optically suppress ACA$_{VIS}$ neuron projection terminals with temporal precision during the 5CSRTT.

(B) Representative images of eNpHR3.0 (green)-expressing ACA neurons and DAPI (blue) in the frontal cortex (left; scale bar, 200 μm) and axon terminals in the VIS (right; scale bar, 100 μm).

(C) Experimental timeline. Mice were trained on the 5CSRTT before virus injection. After 3 weeks to allow maximal virus expression, mice performed 5CSRTT. During the 5CSRTT with a fixed 5-s ITI and pseudorandomized stimulus duration (2.0 or 1.0 s), a 470-nm LED was illuminated during early ITI (−5:-2 s), late ITI (−3:0 s), or during the period of stimulus presentation (0:1 s) in 50% of the trials (8 mice, 3,525 total trials). One timing was tested per day. The effect of optogenetic suppression during early ITI, late ITI, or the stimulus period on behavioral performance (correct trial [percent] for a 1-s stimulus) was compared following correct trials (C+1) and following error trials (E+1) at each tested light timing.

(D) Continuous 470-nm illumination during ITI −5:−2 s had no effect on correct trials (percent) after a correct or error trial (general linear mixed model, effect of treatment [light on versus off]: p = 0.8802).

(E) Change in correct trials (percent light on – light off) during a −5:−2 s period was not different from zero in any group (Wilcoxon signed-rank test; C+1: p = 0.3750, E+1: p = 0.8125).

(F) Continuous 470-nm illumination during ITI −3:0 s had no effect on correct trials (percent) after a correct trial but decreased performance after error trials (general linear mixed model, effect of treatment [light on versus off]: p = 0.0032, planned Tukey post hoc tests: C+1 light off versus on: p = 0.8976, E+1 light off versus on: **p = 0.0054).

(G) Change in correct trials (percent light on – light off) during ITI −3:0 s stimulation was significantly different from zero only in the E+1 group (Wilcoxon signed-rank test, C+1: p = 0.2969, E+1: *p = 0.0234).

(H) Continuous 470-nm illumination during the stimulus period 0:1 s had no effect on correct trials (percent) after a correct or error trial (general linear mixed model, effect of treatment [light on versus off]: p = 0.7054).

(I) Change in correct (percent light on – light off) during the visual stimulus period was not significantly different from zero in any group (Wilcoxon signed-rank test; C+1: p = 0.2500, E+1: p = 0.0547).

Error bars indicate mean ± SEM. *p < 0.05, **p < 0.01. See also Figures S6 and S7.

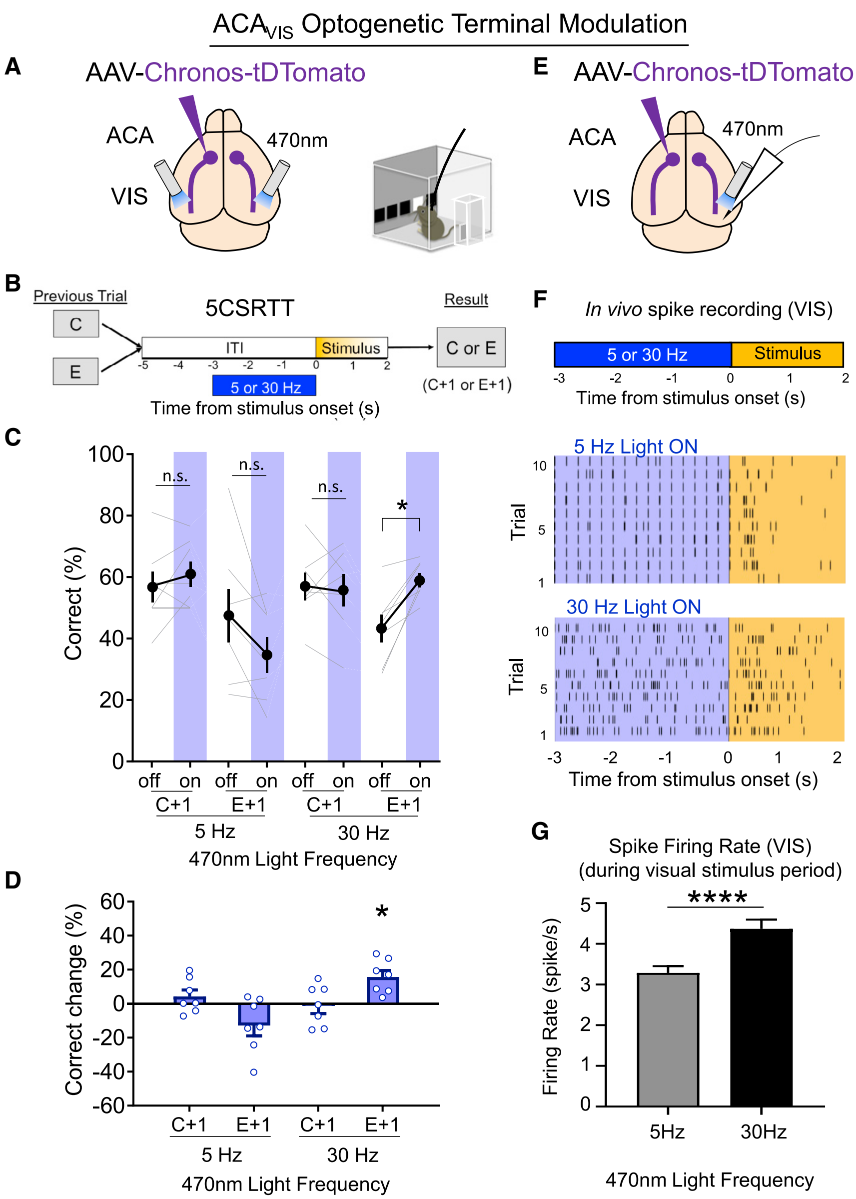

Figure 5. 30-Hz optogenetic stimulation of ACA$_{VIS}$ neuron projection terminals in the VIS promotes visual processing and post-error performance

(A–D) Optogenetic ACA$_{VIS}$ neuron terminal stimulation during the 5CSRTT.

(A) Viral strategy. To express light-gated channel Chronos in ACA neurons, Chronos encoding AAV was injected into the ACA. An optic fiber was implanted at the VIS to optically modulate ACA$_{VIS}$ neuron projection terminals.

(B) Experimental design. 470-nm light was illuminated at 5 or 30 Hz during the last 3 s of the ITI (−3:0 s) in 50% of trials (7 mice, 1,929 total trials).

(C) 30-Hz stimulation during the ITI showed a significant increase in correct trials (percent) after an error but not after a correct response. However, 5-Hz stimulation during the ITI had no effect on correct trials (percent) after an error or after a correct response (general linear mixed model, effect of treatment [light on versus off] p = 0.6013, effect of frequency [5 versus 30 Hz] p = 0.0659, effect of trial type [C+1 versus E+1] p < 0.0001, planned Tukey post hoc tests: 5 Hz:C+1 light off versus on: p = 1.000, E+1 light off versus on: p = 0.4026, 30 Hz: C+1 light off versus on: p = 1.000, E+1 light off versus on: *p = 0.0289).

(D) Change in correct trials (percent light on – light off) during ITI stimulation was significantly different from zero only in the 30 Hz:E+1 group (Wilcoxon signed-rank test; 5 Hz:C+1: p = 0.5625, 5 Hz:E+1: p = 0.1563, 30 Hz:C+1: p = 0.8125, 30 Hz:E+1: *p = 0.0156).

(E–G) Visually evoked spike activity in the VIS following optogenetic modulation of ACA$_{VIS}$ neuron terminals in anesthetized mice.

(E) Experimental design. *In vivo* extracellular single-unit recordings of VIS neurons of anesthetized mice during and immediately after blue 470-nm light illumination at 5 Hz or 30 Hz over the VIS to optogenetically activate ACA$_{VIS}$ neuron projection terminals in mice virally expressing Chronos in ACA neurons.

(F) Representative spike activity of VIS neurons upon 5-Hz (top) or 30-Hz (bottom) blue light illumination, shown as raster plot and firing rates (FRs) before, during (ON: −3:0 s), and after (post: 0:2 s).

(G) Quantification of VIS FRs after blue light illumination at 5 Hz or 30 Hz (unpaired t test, t_{738} = 3.945, ****p < 0.0001, n = 402 cells [5 Hz] and 338 cells [30 Hz] from 5 mice). Error bars indicate mean ± SEM.

*p < 0.05, ****p < 0.0001. See also Figures S8 and S9.

mPFC PV interneurons reduced total errors and improved attention performance (Kim et al., 2016). As rats performed a similar attention task, 13- to 30-Hz LFP power in the ACA increased during stimulus anticipation in correct compared with incorrect trials (Totah et al., 2013). Together with previous findings, our study suggests that the temporal organization needed for effective ACA$_{VIS}$ circuit activity may constrain when optogenetic stimulation does or does not improve performance. The converging methods used in this study (i.e., optogenetic manipulation during behavior, EEG during behavior, and optogenetics in anesthetized mice) support a model where specific patterns of activity in the ACA$_{VIS}$ pathway on VIS activity are critical components of post-error attention mechanisms. The current study is limited, however, because we assessed the effects of optogenetic stimulation on spike and LFP activity in anesthetized rather than behaving mice. To better establish a direct link between ACA$_{VIS}$ projections, network activity, and behavior, when technological innovations allow such work, future studies should incorporate single-unit recordings and optogenetics simultaneously as animals perform attention tasks.

Our study also provided mechanistic insights into how optogenetic stimulation prior to visual stimulus presentation can influence visual attention. *In vivo* recordings in anesthetized mice showed that 30-Hz optogenetic stimulation of ACA_{VIS} neuron terminals increased spiking in the VIS in response to subsequent visual stimulus presentation, a key hallmark neural correlate of attention (Noudoost et al., 2010). Our results are consistent with the increased probability of spiking that accompanies gamma oscillations (Buzsáki and Watson, 2012; Salinas and Sejnowski, 2000, 2001). The oscillatory activity established by optical stimulation immediately prior to onset of visual stimuli may promote subsequent resonance—temporally coordinated oscillations in the dendrites of local circuits that reduce the threshold for stimulus detection upon stimulus presentation without interference. Our findings support a model where optogenetic stimulation of ACA_{VIS} neurons by 30-Hz light illumination evokes a selective increase in gamma oscillations, which, in turn, may improve subsequent visual processing after an error to facilitate task performance. Future studies are warranted to directly test this model by incorporating single-unit recordings and optogenetics simultaneously as animals perform attention tasks.

Optogenetic stimulation during visual stimulus presentation did not improve task performance regardless of prior trial outcomes, whereas ACA_{VIS} neuron activity increased during the ITI and stimulus presentation after error (Figure 1E). A previous study also showed that activating ACA_{VIS} neuron terminals during stimulus presentation increased firing rates in a subset of orientation-sensitive neurons (Zhang et al., 2014). In the present experiment, stimulation during the presentation of visual stimuli did not improve behavioral performance, perhaps because the exogenous optogenetic stimulation interfered with the endogenous timing of activity patterns needed for processing large-field visual stimuli (e.g., (Knudsen and Wallis, 2020). The non-constrained experimental setting of 5CSRTT, which engages additional behavioral mechanisms, such as head orientation to adjust behavioral performance, may also have contributed to masking the beneficial effect of optogenetic stimulation on task performance.

Deficits in error monitoring (Kerns et al., 2005; van Veen and Carter, 2006) and attention network function (Mazaheri et al., 2010; Roiser et al., 2013) are common to many psychiatric disorders, including ADHD, schizophrenia, and autism spectrum disorder. The link between these cognitive deficits and distinct neural circuits remains unclear. Our study suggests that modulating activity in specific cortical pathways could provide a means for improving post-error attention in psychiatric disorders (Michelini et al., 2016).

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
  - Lead contact
  - Materials availability
  - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
- METHOD DETAILS
  - Viral Strategies and Stereotaxic Procedures
  - Behavior
  - *In vivo* Fiber Photometry
  - EEG Recordings
  - Optogenetics
  - *In vivo* electrophysiology
  - Slice electrophysiology
  - Immunohistochemistry
  - Imaging and analysis
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.neuron.2021.02.001.

ACKNOWLEDGMENTS

This work was funded by NIH F31MH121010 (to K.J.N.); NIH F30MH111143; the Beatrice and Samuel A. Seaver Foundation (to E.N.F.); R01MH073689 (to M.L.S.); and NIH R21NS105119, R21MH106919, R01EY024918, and R01MH119523 (to H.M.). We thank Dr. Erin Rich, Nick Upright, and members of the Morishita lab for helpful feedback and the NVIDIA Corporation.

AUTHOR CONTRIBUTIONS

K.J.N., H.K., and H.M. designed the experiments. K.J.N. and H.M. performed analyses and wrote the manuscript with contributions from all co-authors. K.J.N. performed fiber photometry and optogenetics experiments with assistance from H.K., J.B., S.E.M., K.C., C.C., E.N.F., M.P.D., P.M., and L.W. K. Yamamuro performed slice electrophysiology. A.L., D.K., M.S., and K. Yoshitake performed *in vivo* electrophysiology experiments. J.S.R. and M.L.S. analyzed spike/LFP/EEG data, and J.B. and P.H.R. assisted with analysis. M.E.F. and S.J.R. provided expertise regarding photometry methodology. K.K. and A.W.V. provided expertise regarding EEG recording methodology. D.M.B. wrote the Python script for fiber photometry analysis. M.G.B. provided guidance regarding behavioral experiments and statistical analysis. H.M. supervised the research.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in science. One or more of the authors of this paper self-identifies as a member of the LGBTQ+ community. One or more of the authors of this paper self-identifies as living with a disability. While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list. The author list of this paper includes contributors from the location where the research was conducted who participated in the data collection, design, analysis, and/or interpretation of the work.

Received: April 9, 2020
Revised: October 26, 2020
Accepted: January 29, 2021
Published: February 19, 2021

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
rabbit anti-GFP antibody	Invitrogen	A-11122; RRID:AB_221569
Alexa 488-conjugated goat anti-rabbit IgG	Invitrogen	A-11034; RRID:AB_2576217
Bacterial and virus strains		
AAV1-Syn-FLEX-GCaMP6f-WPRE-SV40	Chen et al., 2013	Addgene viral prep # 100833-AAV1
AAV1-EF1a-doublefloxed-hChR2(H134R)-eYFP-WPRE	A gift from Karl Deisseroth	Addgene viral prep # 20298-AAV1
AAV2-CamKII-eNphR3.0-eYFP-WPRE-PA	UNC vector core	N/A
AAV2-CamKII-GFP	UNC vector core	N/A
AAV8-Syn-Chronos-tdTomato	Klapoetke et al., 2014	Addgene viral prep # 62726-AAV8
rAAV2-CAG-CRE-WPRE	Boston Children's Hospital vector core	N/A
Experimental models: organisms/strains		
Mouse:C57BL/6	Charles River Laboratories	N/A
Software and algorithms		
MATLAB	Mathworks	https://www.mathworks.com/products/matlab.html
Sirenia Acquisition software	Pinnacle Technologies	https://www.pinnaclet.com/sirenia-download.html
ImageJ	NIH	https://imagej.nih.gov/ij/index.html; RRID:SCR_003070
Prism8	GraphPad	https://www.graphpad.com
Synapse	Tucker-Davis Technologies	N/A
R	N/A	https://www.r-project.org/
pClamp 10 v. 10.6.2.2	Molecular Devices	N/A
Other		
Nanoinjector	World Precision Instruments	Cat# Nanoliter 2000
Superfrost Plus slides	Fisher Scientific	Cat# 22-037-246
Touch screen chambers for electrophysiology	Lafayette Instruments	Model 80614E
Teleopto wireless optogenetics	Amuza	N/A
digital fiber photometry processor	Tucker-Davis Technologies	RZ5P
OmniPlex Neural Recording Data Acquisition System A	Plexon	N/A
Multiclamp 700B	Axon Instruments	N/A

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Hirofumi Morishita (hirofumi.morishita@mssm.edu).

Materials availability

This study did not generate unique reagents.

Data and code availability

The datasets and the code that support the findings of this study are available from the Lead Contact upon reasonable request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Adult, male C57BL/6 mice (Charles River Laboratories, MA) were group-housed under a standard 12 hour light/dark cycle in a temperature- and humidity-controlled vivarium. Training was initiated when mice were 9–10 weeks old. Mice were allowed access to water for 2 hours each day and maintained approximately 85%–90% of their *ad libitum* weight during behavioral training. Food was available *ad libitum* throughout the experiment. All animal protocols were approved by the Institutional Animal Care and Use Committee at Icahn School of Medicine at Mount Sinai.

METHOD DETAILS

Viral Strategies and Stereotaxic Procedures

Following procedures previously described in Nabel et al. (2020), mice were anesthetized with 2% isoflurane and head-fixed in a mouse stereotaxic apparatus (Narishige, East Meadow, NY). Bilateral ACA injection sites relative to Bregma area are: AP +0.7mm, ML ± 0.2mm, DV −0.7mm; AP +0.2mm, ML ± 0.2mm, DV −0.7mm; AP −0.3mm, ML ± 0.2mm, DV −0.7mm. Bilateral VIS injection sites relative to lambda are: AP +0.0mm, ML ± 3.0mm, DV −0.4mm; AP +0.1mm, ML ± 2.85mm, DV −0.4mm; AP +0.1mm, ML ± 3.15mm, DV −0.4mm. For circuit-specific *in vivo* imaging of ACA_{VIS} neurons, AAV1-Syn-FLEX-GCaMP6f (Addgene) was injected to the ACA and concurrently, retroAAV2-Cre was injected into VIS, unilaterally. A 1.2 mm length fiber optic ferrule was a 0.48 numerical aperture and a 400 μm core diameter (Doric Lenses, Quebec, Canada) was implanted at the center of ACA injection sites and cemented to the skull using dental cement (C&B Metabond, Parkell, Edgewood, NY; Jet Acrylic, Lang Dental, Wheeling, IL). To optogenetically activate ACA_{VIS} projection neurons at ACA soma, AAV1-EF1a-DIO-hChR2-H134R-eYFP (Addgene) and RetroAAV2-Cre (Boston Children's Hospital Viral Core) were injected bilaterally into the ACA and VIS, respectively. Bilateral LEDs (Amuza, San Diego, CA) of 500 μm diameter that delivered 470nm light were implanted at the ACA at the center of injection sites. To optogenetically suppress ACA_{VIS} neurons at terminals in the VIS, AAV2-CamKII-eNphR3.0-eYFP (UNC Viral Vector Core, Chapel Hill, NC) was injected bilaterally into the ACA. AAV2-CamKII-GFP (UNC Viral Vector Core, Chapel Hill, NC) was injected bilaterally into the ACA as static fluorophore control. To optogenetically modulate ACA_{VIS} neurons at terminals in the VIS, AAV8-Syn-Chronos-tdTomato (Addgene) was injected bilaterally into the ACA. For eNpHR3.0 and Chronos experiments, bilateral LEDs (Amuza, San Diego, CA) of 500 μm diameter that delivered 470nm light were implanted at the VIS. Each infusion (500 nl) was made at 150 nl/min using a microinjector set (Nanoject III) and glass pulled syringe. The syringe was left in place for 1min following the injection to reduce backflow of virus. Behavioral testing occurred at least three weeks after viral injection to allow for maximal viral expression.

Behavior

5-Choice Serial Reaction Time Task (5CSRTT):

Apparatus. 5CSRTT (See also Figure S1) was conducted in eight Bussey–Saksida operant chambers with a touchscreen system (Lafayette Instruments, Lafayette, IL) following procedures previously described in Nabel et al. (2020). Dimensions are as follows: a black plastic trapezoid (walls 20 cm high × 18 cm wide (at screen-magazine) × 24 cm wide (at screen) × 6 cm wide (at magazine)). Stimuli were displayed on a touch-sensitive screen (12.1 inch, screen resolution 600 × 800) divided into five response windows by a black plastic mask (4.0 × 4.0 cm, positioned centrally with windows spaced 1.0 cm apart, 1.5 cm above the floor) fitted in front of the touchscreen. Schedules were designed and data were collected and analyzed using ABET II Touch software (v18.04.17, Lafayette Instrument). The inputs and outputs of the multiple chambers were controlled by WhiskerServer software (v4.7.7, Lafayette Instrument).

Habituation. Before 5CSRTT training, mice were first acclimated to the operant chamber and milk reward. The food magazine was illuminated and diluted (30%) sweetened condensed milk (Eagle Brand, Borden, Richfield, OH) was dispensed every 40 s after mice entered the food magazine. Mice needed to enter the reward tray 20 times during two consecutive 30min sessions before advancing to the next stage. Mice were then trained to touch the illuminated response window. During this phase a white square stimulus was presented randomly at one response window until it was touched. If the mouse touched the stimulus, the milk reward was delivered in conjunction with a tone and magazine light. Touches to nonstimuli had no consequence. After reaching criterion on this phase (20 stimulus touches in 30 min for 2 consecutive days), mice advanced to the 5CSRTT training phase.

Training. Mice were tested 5 days a week, 100 trials a day (or up to 30min). Each trial began with the illumination of the magazine light. After mice exited the food magazine, there was an intertrial interval (ITI) period of 5 s before a stimulus was presented at one response window. If a mouse touched the screen during the ITI period, the response was recorded as premature and the mouse was punished with a 5 s time-out (house light on). After the time-out period, the magazine light illumination and house light switch off signaled onset of the next trial. After the ITI period, a stimulus appeared randomly in one of the five response windows for a set stimulus duration (this varied from 32 to 2 s, depending on stage of training). A limited-hold period followed by the stimulus duration was 5 s, during which the stimulus was absent but the mouse was still able to respond to the location. Responses during stimulus presence and limited holding period could be recorded either as correct (touching the stimulus window) or incorrect (touching any other windows). A correct response was rewarded with a tone, and milk delivery, indicated by the illumination of the magazine light. Failure to respond to any window over the stimulus and limited-hold period was counted as an omission. Incorrect responses and omissions were punished with a 5 s time-out. In addition, repeated screen touches after a correct or incorrect response were counted as perseverative responses. Animals started at stimulus duration of 32 s. With a goal to baseline mice at a stimulus duration of 2 s, the stimulus duration was sequentially reduced from 32, 16, 8, 4, to 2 s. Animals had to reach a criterion (> 50 trials, > 80% accuracy, < 20% omissions) over 2 consecutive days to pass from one stage to the next. After reaching baseline criterion with the 2 s stimulus duration for 4 out of 5 days, mice began 5CSRTT testing.

5CSRTT testing. During 5CSRTT testing, attention demand was increased by reducing and pseudo-randomly shuffling the stimulus duration to 2.0, 1.5, 1.0, and 0.8 s for 4 out of 5 days. Following 5CSRTT testing, mice underwent stereotaxic viral surgery and then were reestablished to baseline criterion before fiber photometry imaging (2.0, 1.5, 1.0, and 0.8 s stimulus duration), or optogenetics (2.0 and 1.0 s simulus duration) experiments. Between experimental testing days, mice were subjected to 2 s stimulus duration training to confirm that the mice maintain stable baseline criterion. Attention and response control were assessed by measuring the following performance: correct percentage ((100 x (correct responses)/(correct responses + incorrect responses + omissions)), percentage accuracy (100 × correct responses/(correct responses + incorrect responses)), percentage omission (100 × omissions/(omissions + correct responses + incorrect responses)), percentage of premature responses, percentage of perseverative responses, latency to correct response (s), and latency to reward collection (s) after correct choices. A premature response is a response given during the ITI before a stimulus appears. It is a measure of impulsive behavior. A perseverative response is when a response to any screen continues to be given after the correct or incorrect screen has been poked and before retrieval of the reward. It is a measure of compulsive behavior. Correct response latency (CRL) is the difference between the time of stimulus onset and the time of the correct response. It is a measure of attention and processing speed. Reward collection latency (RCL), is the difference between the time of a correct response and the time of reward collection and reflects motivation.

Progressive ratio task

Following 5CSRTT testing, mice began training for a progressive ratio task following procedures previously described in (Nabel et al., 2020). Training and testing were performed in the Bussey–Saksida chamber as described above. Mice were water-restricted similar to during 5CSRTT. First, mice were subjected to fixed ratio (FR) training in which the center response window was illuminated and needed to be touched for a fixed number of times, depending on schedule, in order for a milk reward to be dispensed. Mice were trained on FR1, followed by FR2, FR3, and FR5 schedules. Criterion was defined as completion of 30 trials in a single session. After passed the criterion on FR5 phase for three consecutive days, mice were subjected to the progressive ratio (PR) phase where the reward response requirement was incremented on a linear +4 basis (i.e., 1, 5, 9, 13 etc.). Once mice showed stable performances, defined as less than 10% breakpoint variability across two sessions, mice underwent four days of testing. For excitatory optogenetics experiments, LEDs were stimulated at 30Hz, 1.66ms pulse, 5% duty cycle) for 5 s prior to presentation of illumination of the center response window for each trial of the session. For inhibitory optogenetics experiments, LED was turned on for 3 s continuously prior to presentation of illumination of the center response window for each trial of the session. The order of LED ON versus OFF daily sessions were counterbalanced across mice. Progressive ratio testing was used to measure motivation.

Open field test

Locomotor activity was measured for 30min in a square apparatus (43 cm × 43 cm × 33 cm) equipped with a panel of infrared beams (16 beams) located in the horizontal direction along the sides of each square apparatus. Data were collected with Fusion v4 software (Omnitech Electronics, Columbus, OH, USA). Mice were given *ad libitum* access to water for at least one week prior to open field testing. For excitatory optogenetics experiments, mice underwent two days of open field testing. During tested LED was either turned on at 30Hz (5 s on, 5 s off) for 5min followed by 5min of Light OFF for a total of 30min, order was counterbalanced between mice. Total distance moved (cm) was compared between Light ON and Light OFF 5min epochs.

In vivo Fiber Photometry

Experiments were done at least three weeks after viral injection to allow for sufficient viral expression. A mono-fiber optic cannula (MFC 400/430-0.48 1.2 MF2.5 FLT, Doric Lenses) was stereotaxically implanted above the ACA. During recording, a matching fiber optic patch cord (Doric Lenses, MFP_400/430/0.48_1.3) was attached to the implanted fiber optic cannula with cubic zirconia sheathes following procedures previously described in Bicks et al. (2020). The fiber optic cable was coupled to two excitation LEDs (Thorlabs, Newton, NJ): 470nm to stimulate GCaMP6f fluorescence in a Ca^{2+}-dependent manner and 405nm which serves as an isosbestic wavelength for GCaMP6f, for ratiometric measurements of GCaMP6f activity, correcting for bleaching and signal artifacts. 470nm excitation was sinusoidally-modulated at 210 Hz and 405nm excitation was modulated at 330 Hz. Light then passed through a GFP (490) or violet (405) excitation filter (Thor labs) and dichroic mirrors before passing through the fiber optic cannula into the brain. GFP excitations emissions passed through the fiber optic cable is passed through the dichroic mirror and emission filters, and through a 0.50 N.A. microscope lens (62-561, Edmund Optics, Barrington, NJ) and then focused and projected onto a photodetector (Model 2151 Femtowatt Photoreceiver, Newport Corporation, Irvine, CA). A digital fiber photometry processor (RZ5P, Tucker-Davis Technologies, Alachua, FL) was used to demodulate the signal from each LED. Signal was then pre-processed using the Synapse Software Suite (Tucker-Davis Technologies) and collected at a sampling frequency of 1018 Hz. The power at the fiber-optic tip was approximately 10 mW. 405 nm signal was used to detrend 470 nm based on signal bleaching. The $\Delta F/F$ was then calculated as: (470 nm signals – median 470 nm signals) / (median 470 nm signals) based on each individual 30min 5CSRTT session. Fiber photometry signal data was divided into bins to assess differences in ACA_{VIS} recruitment depending on overall performance (correct, incorrect, or omission), CRL, incorrect response latency (IRL), and RCL. Correct and error trials were plotted to show differences in activity based on overall performance. IRL, CRL, and RCL trials divided into slow and fast responses on a within-mouse basis. Each mouse performed two days of testing. Both days of testing were combined, and then data was split into slow and fast behavior bins based on the median CRL, IRL, and RCL value.

EEG Recordings

Mice were implanted with skull screws located at VIS (relative to lambda: AP +0.0mm, ML ± 3.0mm) to record cortical EEG (Pinnacle, USA). Mice were then allowed 7 days to recover before initiating retraining. Skull screws were implanted at the left and right cerebellum to serve as ground and common reference for recording EEG. EEG signals were captured via a tethered system (Pinnacle, USA) and acquired using Sirenia Acquisition software. Data were acquired at 2 kHz using a Pinnacle preamplifier with a high (0.5 Hz)- and low-pass (200 Hz) filters. Signals were segmented into single trial epochs aligned to stimulus-onset. For time-frequency analyses, signals were padded with a time-reversed copy of themselves to prevent wrap-around effects due to convolution in the frequency domain and wavelet transformed using Morlet wavelets between 4-50hz (Guise and Shapiro, 2017). The wavelet factor (i.e., the parameter controlling the tradeoff between frequency and time resolution) linearly scaled between 14 and 24. For analyzing targeted frequency bands theta (4-8hz) and gamma (25-50hz), signals were Butterworth filtered and Hilbert transformed to generate the analytic signal from which power was extracted. All analyses used custom-written MATLAB routines.

Optogenetics

TTL signals were to a signal generator (Rigol Technologies, Beijing, China) that drove the light at specific pulse frequencies (5Hz-10ms pulses, 30Hz- 1.66ms pulses; 5% duty cycle) during the 5 s ITI or stimulus presentation pseudo randomly during 50% of trials for ChR2 experiment or during the last 3 s of the ITI for Chronos experiments. One frequency was tested per testing day in a counterbalanced manner. For *in vivo* optogenetic inactivation of ACA_{VIS} neuron projections during 5CSRTT, continuous 470nm light was delivered bilaterally via LED optic fiber implanted at the VIS pseudo randomly at specific time points throughout a trial during 50% of trials. In our study, we chose to use 470nm LED to suppress eNpHR3.0-expressing ACA neurons, because previous studies found that red-shifted light frequencies resulted in off-target effects in control mice that reduced 5CSRTT performance (White et al., 2018). During separate test sessions, LEDs were stimulated during either −5:-2 s of ITI, −3:0 s of ITI or during the length of the stimulus presentation. One timing of stimulus duration was used per test day in a counterbalanced manner. During optogenetics testing, stimulus duration was pseudo randomly shuffled between 2.0 and 1.0 s.

In vivo electrophysiology

Three weeks after virus injection, electrophysiological experiments with optogenetic stimulation were performed following procedures previously described in Nabel et al. (2020). For recordings from mice expressing ChR2 in ACA_{VIS} neurons, recordings were performed under 0.8%–1% isoflurane anesthesia. For recordings from mice expressing Chronos in ACA neurons, preparatory surgery leading to, and recording itself was performed initially under Nembutal/chlorprothixene anesthesia and then maintained with isoflurane (Sadahiro et al., 2020). Sixteen-channel silicon probes with 177 μm^2 recording sites (NeuroNexus Technologies, Ann Arbor, MI) spaced 50 μm apart were used to record neuronal activity in the visual cortex (AP +0.0mm, ML +2.80∼+3.2 mm, DV −1.0 mm). The signal detected from the probe was amplified and thresholded, and unit-sorted using an OmniPlex Neural Recording Data Acquisition System A (Plexon). Sorting of single units was carried out using principal component analysis in an offline sorter v. 3.2.2 (Plexon). Spike signals were filtered at a bandpass of 300 Hz to 8 kHz, and LFPs were filtered at a bandpass of 200 Hz. A template online sorting method was used to capture spikes as units. The waveforms of recorded units were further examined using Offline Sorter (Plexon). Blue light illumination (wavelength 473 nm, 5 Hz (10 ms pulses) or 30 Hz (1.66 ms pulses)) was delivered using an optic fiber (diameter 105 μm) and oriented immediately above the ACA (AP −0.3 mm, 0.0 mm, +0.3 mm, +0.6 mm, ML ± 0.25 mm) for ChR2 characterization, or the VIS (AP +0.0mm, ML +2.80∼+3.2mm) cortical surface for Chronos characterization. The power at the fiber-optic tip was approximately 10 mW. For ChR2-related experiments, each recording consists of 5 s of blue light illumination at either 5Hz or 30Hz pulses with more than 1 min intervals between illuminations repeated 10 times. For Chronos-related experiments, 3 s of blue light illumination at 5Hz or 30Hz pulses immediately followed by a presentation of visual stimulus (50% contrast white square (1400pix X 1400pix) in black background (2048pix X 1536pix) with CRT monitor 40.8cm x 30.6cm) programed by Psychopy3 v2020.1.3 was repeated 10 times with 20 s interval. A monitor was presented 15 cm in front of the subject. To assess if laser pulses influence spiking activity, spike modulation indices and probabilities were calculated for each unit. Modulation indices measured the extent to which spiking activity changed during optogenetic stimulation, compared to the 3 s prior to optogenetic stimulation: (ON-pre)/(ON+pre). Light-induced spike probability expresses the ratio between spikes fired and the number of pulses during laser ON. For LFP analysis, signals were first segmented into single trial epochs aligned to light-offset, and then padded with a time-reversed copy of themselves to prevent wrap-around effects due to convolution in the frequency domain and wavelet transformed using Morlet wavelets between 4-50hz (Guise and Shapiro, 2017). The wavelet factor (i.e., the parameter controlling the tradeoff between frequency and time resolution) linearly scaled between 14 and 24. For analyzing targeted frequency bands theta (4-8hz) and gamma (25-50hz), signals were Butterworth filtered and Hilbert transformed to generate the analytic signal from which power was extracted. All analyses used custom-written MATLAB routines.

Slice electrophysiology

Following procedures previously described in Bicks et al., 2020, animals were decapitated under isofluorane anesthesia. Brains were quickly removed and transferred into ice-cold artificial cerebrospinal fluid (ACSF) of the following composition (in mM): 210.3 sucrose, 11 glucose, 2.5 KCl, 1 NaH_2PO_4, 26.2 $NaHCO_3$, 0.5 $CaCl_2$, and 4 $MgCl_2$. Acute coronal slices of ACA (300 μm) contained both hemispheres. Slices were allowed to recover for 40 min at room temperature in the same solution, but with reduced sucrose (105.2 mM) and addition of NaCl (59.5 mM). Following recovery, slices were maintained at room temperature in standard ACSF composed of the following (in mM): 119 NaCl, 2.5 KCl, 1 NaH_2PO_4, 26.2 $NaHCO_3$, 11 glucose, 2 $CaCl_2$, and 2 $MgCl_2$. Slices were visualized under upright differential interference contrast microscope equipped with high power LED-coupled light source used for the identification of fluorescently-labeled cells and optogenetic stimulation (Prizmatix). Patch clamp recordings were made from fluorescently labeled ACA neurons. Borosilicate glass electrodes (5–7 MΩ) were filled with the internal solution containing (in mM): 127.5 K-methanesulfonate, 10 HEPES, 5 KCl, 5 Na-phosphocreatine, 2 $MgCl_2$, 2 Mg-ATP, 0.6 EGTA, and 0.3 Na-GTP (pH 7.25, 295 mOsm). Signals were low-pass filtered at 3 kHz and sampled at 20 kHz. Data were low-pass filtered at 3 kHz and acquired at 10 kHz using Multiclamp 700B (Axon Instruments) and pClamp 10 v. 10.6.2.2 (Molecular Devices). Neurons were included in the analysis if input resistance, series resistance and membrane potential did not change more than 10% during the course of recordings. To validate modulation of ACA neurons with eNpHR3.0, whole-cell recordings in gap-free mode were obtained with borosilicate glass electrodes (5-8 mΩ resistance) filled with a current clamp internal solution. Cells were held at −55mV. Frequency of spikes induced by current injection in neurons expressing eNpHR3.0 were compared before, during, and after blue light stimulation using TTL-pulsed microscope-coupled LED (470 nm, Prizmatix, Holon, Israel).

Immunohistochemistry

After mice completed behavior training and testing, they were anesthetized with isofluorane and underwent transcardial perfusion with phosphate-buffered saline (PBS) followed by 4% paraformaldehyde (PFA) in PBS, with overnight postfixation. Brains were coronally sectioned on a Leica CM3050 S cryostat (Leica Microsystems, Buffalo Grove, IL) at 30 μm for immunohistochemistry. Slices were placed in a blocking solution of 1% Bovine Serum Albumin (BSA) with 0.1% Triton X-100 in PBS on a rotating shaker for 1 hour at room temperature (RT). The slices were incubated overnight on a rotating shaker at RT in primary antibody rabbit anti-GFP (Invitrogen, 1:1000). After the overnight incubation, slices were washed three times in blocking solution then incubated for 2.5 hours at RT in secondary antibody 488 donkey anti-rabbit (Invitrogen, 1:400). The slices were subsequently washed three times, twice in blocking solution and finally in PBS. Slices were mounted on microscope slides (Brain Research Laboratories, Waban, MA) using DAPI Fluoromount-G mounting medium (Southern Biotech) and coverslipped. Using the Allen Brain Atlas, slices were collected at nine specific Bregma areas (0.14, −0.34, −0.82) to analyze the anterior-posterior spread of virally infected cells. Visual cortex slices were selected at seven Bregma areas (−3.58, −3.98, −4.38).

Imaging and analysis

Images to validate the cannula location and viral spread of the GCaMP, ChR2, eNpHR3.0, GFP, and Chronos groups were acquired using an EVOS FL Imaging System (Thermo Fisher Scientific, Waltham, MA). Slices were imaged using a 10x lens for GFP and DAPI. To ensure consistency across imaging sessions, to image GFP signal, power was set at 90% and to image DAPI, power was set at 20%–30%. Representative images were acquired using an Inverted DMi8 Widefield Microscope (Leica, Microsystems). Slices were imaged using a 20x lens for GFP and DAPI. To ensure consistency across imaging sessions, to image GFP signal, power was set at 70% and exposure time for 500 ms, and to image DAPI, power was set at 50% and exposure time for 250 ms. Images were analyzed using ImageJ software (National Institutes of Health). Mice were excluded if no expression levels were seen or if expression was not present in the dorsal ACA. Optic Fiber location analysis: The location of the cannula was mapped onto blank coronal slice templates taken from Paxinos and Franklin mouse brain atlas. The bottom most edge of the oval indicates how deep into the tissue the cannula was placed. Mice were excluded if cannula location implant was inaccurate.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed using Prism (Graphpad, San Diego, CA) and in R v. 3.5.3. Analyses were conducted using general linear mixed models, ANOVA, Wilcoxon signed-rank tests, or t tests as indicated. Bar graphs and photometry averaged traces are represented as the mean and error bars represent standard error of the mean (s.e.m.)